

High-Speed DWDM Technology

Dense wavelength division multiplexing (DWDM) is a promising transmission technology that can offer very high bandwidth capabilities in the range of terahertz. Such capabilities are possible due to transmission of a large number of high-speed channels simultaneously

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The rapid growth of information technology has led to new services hungry for transmission capacity. Services such as voice-over-Internet protocol (VoIP), video streaming, telemedicine, telecommuting, broadcasting of TV programmes, high-speed file sharing, user-generated video, online video gaming, online education and shopping need a transmission medium with very high bandwidth capabilities for handling the vast amounts of information. Fibre-optics, with its comparatively infinite bandwidth and employing the latest multiplexing techniques, has proven to be the solution.

Multiplexing techniques

Information transfer from one place to another takes place through different types of media such as copper lines, fibre-optic cables and space. To exploit maximum information-carrying capabilities of these transmission media, several information channels are simultaneously sent over the common transmission medium. This technique is known as multiplexing.

In time-division multiplex access (TDMA), time on the transmission medium is shared amongst multiple data sources, that is, all signals operate with the same frequency at different times. It is a digital technique that divides a single channel or band into time slots (Fig. 1). Each time slot is used to transmit one byte or another digital segment of each signal in sequential serial data format. An example of TDMA-based technology is the basic global system for mobile communications (GSM).

A variant of TDMA, known as statistical/asynchronous/intelligent

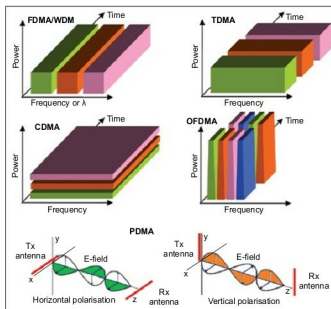


Fig. 1. Different multiplexing techniques

TDMA, dynamically allocates the time slots, on demand, to separate input channels, thus saving the channel capacity. In case of statistical TDMA, the data in each slot must have an address part which identifies the source of data.

In frequency division multiplex access (FDMA), all signals operate at the same time with different frequencies. Thus, it is a process of dividing one channel or bandwidth into multiple individual bands, each for use by a single user (Fig. 1). The information to be transmitted is modulated on to each subcarrier, and all this is linearly mixed together. The best example of FDMA-based system is the cable television system.

Orthogonal frequency-division multiplexing (OFDM)—a variant of FDM with more efficient spectral efficiency—distributes the data over a large number of carriers that are closely spaced at precise frequencies (until they are actually overlapping). This is done by finding frequencies that are

orthogonal, which means that they are perpendicular in a mathematical sense, allowing the spectrum of each sub-channel to overlap another without interfering (Fig. 1). Worldwide interoperability for microwave access (WiMAX) and long-term evolution (LTE) cellular systems are good examples using OFDM.

Code-division multiple access (CDMA) is a radio technology in which a spread spectrum signal is generated by modulating the radio

frequency (RF) signal with a code comprising different pseudorandom binary sequences for each user, which is inherently resistant to a noisy signal environment.

A number of spread spectrum RF signals thus generated share the same frequency spectrum. Thus the entire bandwidth available in the band is used by each of the users using the same frequency at the same time (Fig. 1). It is a pure digital technology in which information is digitised, compressed and processed in a logic circuit with a higher frequency coded chipping signal.

The chipping signal is derived from a pseudorandom code generator that assigns a unique code to each user of the channel. This code spreads the information signal over a wide bandwidth. The resulting signal is at a low power level and appears more like noise. Many such signals can occupy the same channel simultaneously. The third generation (3G) cell phone technology called wide-band CDMA (WCDMA) uses a similar method.